

CHRONUS

A Memory-Trained Agentic AI for Human Personality and Mindset Simulation

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Submitted by

TIRTH SAKARIYA (2303031050723)

MANAV AGHERA (2403031057168)

MANISH PRAJAPATI (2303031050721)

Under the Guidance of

SHUBHAM UPADHYAY

PIET, Parul University



Parul® University
NAAC A++

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Certificate

This is to certify that the research work titled '**CHRONUS: A Memory-Trained Agentic AI for Human Personality and Mindset Simulation**' is a bonafide record of work carried out by **Tirth Sakariya, Manav Aghera, and Manish Prajapati** under my guidance at PIET, Parul University, during Academic Year **2025-2026**.

To the best of my knowledge, this work has not been submitted in full or in part for any other academic award and adheres to ethical and technical standards expected of an undergraduate engineering research submission.

Project Guide: **Shubham Upadhyay**

Head of Department: _____

Date of Submission: _____

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We acknowledge the open-source communities and research ecosystems behind Sentence-BERT, ChromaDB, Flask, React, and neural speech technologies, whose foundational work made this implementation feasible. We also thank our families and peers for moral support during the project lifecycle.

Abstract

The loss of human knowledge, personality, and wisdom at the end of life represents a profound information discontinuity. Existing systems preserve fragments such as text archives or media clips, but rarely reconstruct a queryable, grounded, and privacy-conscious cognitive representation. This paper presents CHRONUS, a memory-trained agentic AI system designed to simulate mindset and personality using personal records, structured interviews, and semantic retrieval.

The implemented prototype ingests personal artifacts through a Python pipeline, segments content into memory units, encodes them with all-MiniLM-L6-v2 sentence embeddings, and persists vectors in a local ChromaDB instance. At inference time, user questions are embedded and matched to memory passages through similarity search, then rendered as grounded persona responses. The user experience is delivered through a React + TypeScript frontend and a Node/Express proxy integrated with a Flask backend.

The system emphasizes factual grounding and local-first privacy. Retrieval-centric response generation minimizes hallucination compared with unconstrained role-play prompting. The current repository demonstrates practical viability of semantic memory chat for grief-support and digital-legacy contexts and provides a measurable path toward richer capabilities such as robust voice cloning, multimodal memory ingestion, and stricter safety governance.

Keywords: Agentic AI, Memory Simulation, Sentence-BERT, ChromaDB, Persona Modelling, Digital Legacy, Grief Technology, Retrieval Grounding.

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Chapter 1: Introduction

Human identity is more than isolated facts; it is the continuity of memory, values, language habits, and emotional framing. When a person is no longer available, this continuity collapses into disconnected artifacts. CHRONUS is built to reduce that discontinuity by transforming scattered personal records into an interactive, queryable memory model that reflects a person's documented mindset.

This research positions CHRONUS at the intersection of AI systems engineering and grief technology. Unlike purely generative companions, the design objective is not unrestricted creativity; it is grounded memory reconstruction with explicit provenance behavior. The practical hypothesis is that semantically retrieved memories generate more trustworthy and emotionally stable interactions than free-form role-play prompts.

1.1 Scope of Work

- Build and test a local-first architecture for memory-grounded chat.
- Support multi-source data ingestion and structured interview augmentation.
- Evaluate retrieval quality and user-perceived conversational authenticity.
- Document ethical boundaries for grief-sensitive deployment.
- Prepare a roadmap from prototype to clinically informed product.

1.2 Aim

The aim is to design a memory-trained agentic platform capable of producing context-aware responses that are semantically grounded in source material while preserving privacy and minimizing fabrication risk.

1.3 Objectives

- Create a robust retrieval core based on sentence embeddings and vector search.
- Integrate the core with a full-stack interactive interface for real users.
- Quantify behavior using retrieval and relevance metrics.
- Analyze deployment risks in bereavement and legacy-preservation contexts.
- Define an implementation path for voice and multimodal extensions.

1.4 Problem Dimensions

The research addresses five dimensions that repeatedly fail in existing solutions:

- Identity loss: documented thought is preserved, but dialog access is missing.
- Archive fragmentation: personal memories are spread across incompatible formats.
- Trust gap: generative systems often produce plausible but incorrect statements.
- Privacy risk: intimate family data is frequently pushed to cloud services.
- Continuity gap: systems rarely support progressive enrichment over time.

1.5 Research Contribution

This paper contributes an end-to-end blueprint and prototype evidence for memory-grounded persona interaction. It details architecture, implementation decisions, operational constraints, and governance practices required to convert an emotionally meaningful idea into an auditable engineering system.

Chapter 2: Literature Review

The literature is reviewed across conversational systems, embedding models, vector retrieval infrastructure, speech technologies, and anti-hallucination strategies. Emphasis is placed on what each stream solves, where it falls short for grief technology, and how those gaps inform CHRONUS design choices.

2.1 Conversational Digital Legacy Systems

Replika introduced personalized companion behavior but depends heavily on generative style fitting, limiting factual grounding.

HereAfter and StoryFile validate legacy-memory interaction, yet depend on pre-recorded assets and constrained query spaces.

Emotionally engaging outputs often come at the cost of verifiable source traceability.

2.2 Transformer-Based Semantic Encoding

BERT created contextual representation depth but is inefficient for corpus-wide pairwise similarity when used naively.

Sentence-BERT enabled practical sentence-level embeddings for scalable retrieval and semantic clustering.

all-MiniLM-L6-v2 offers a favorable quality-latency trade-off for local CPU execution.

2.3 Vector Databases and ANN Search

FAISS and related ANN techniques established scalable nearest-neighbor retrieval in high-dimensional spaces.

ChromaDB simplifies local vector persistence with metadata support and developer-friendly APIs.

Cloud vector stores scale better but conflict with privacy-by-default principles in grief use cases.

2.4 Voice Cloning and TTS

WaveNet and Tacotron families demonstrated near-human speech quality in controlled settings.

Commercial APIs provide fast integration but raise consent, retention, and jurisdiction concerns.

Local TTS models improve data sovereignty at the cost of setup complexity and compute overhead.

2.5 Anti-Hallucination and Grounding

Prompted persona simulation without retrieval is vulnerable to fabrication and narrative drift.

RAG-style systems reduce hallucination by constraining response synthesis with external evidence.

For memory simulation, deterministic or semi-deterministic composition around retrieved evidence can provide stronger integrity guarantees.

2.6 Synthesized Gap Analysis

No single prior system combines semantic retrieval grounding, open-format personal data ingestion, local-first deployment constraints, structured interview augmentation, and an implementation path for identity-consistent audio delivery under explicit consent. CHRONUS targets that integrated gap.

Chapter 3: Problem Statement and Research Questions

The primary problem is preserving and querying a person's cognitive footprint without distorting identity or violating privacy. The engineering challenge is not only generation quality but lifecycle integrity: ingestion, indexing, retrieval, response composition, auditability, and controlled deletion.

3.1 Problem Statement

Given a set of personal documents and optional interview responses, build a system that returns contextually relevant, identity-consistent responses grounded in source memory units while operating locally and providing transparent fallback behavior when confidence is low.

3.2 Research Questions

- RQ1: Can local sentence-embedding retrieval maintain useful conversational relevance for personal memory Q&A?
- RQ2: Which response-composition patterns minimize hallucination while preserving conversational naturalness?
- RQ3: What data volume and diversity are needed for stable persona quality?
- RQ4: How should consent and deletion be operationalized in family-centered memory systems?
- RQ5: What architecture choices support future multimodal expansion without rewriting the core?

3.3 Hypotheses

- H1: Semantic retrieval outperforms keyword-only matching for open-ended personal questions.
- H2: Grounded response templates reduce fabrication relative to unconstrained generation.
- H3: Structured interview data materially improves model stability when documents are sparse.
- H4: Local-first deployment increases user trust for sensitive memory datasets.

3.4 Operational Definitions

For consistent measurement, the following terms are used:

- Memory unit: one retrievable sentence or compact textual segment.

- Grounded response: answer assembled from retrieved memory evidence.
- Persona consistency: stability of tone and values across related questions.
- Fabrication event: claim unsupported by available memory units.
- Privacy boundary: the explicit runtime and storage perimeter for sensitive data.

Chapter 4: System Architecture

Repository analysis shows a layered architecture with a React frontend, Node/Express proxy, Flask service layer, and local ChromaDB-backed memory engine. This architecture balances developer productivity with deployment simplicity on commodity hardware.

4.1 Frontend Layer

The frontend is built with React and TypeScript under Vite, using utility-driven styling and a componentized UI structure. It manages upload workflows, model selection, interview progression, and chat interaction with asynchronous API calls.

4.2 API Proxy Layer

Node/Express serves as a thin orchestration layer, handling app serving and endpoint forwarding. This separation decouples UX deployment from Python inference runtime and allows independent scaling or replacement later.

4.3 Backend Service Layer

Flask endpoints provide health checks, file upload and management, interview orchestration, model training, pretrained model handling, and chat response delivery. The endpoint design supports stepwise user flows and explicit error surfaces.

4.4 Memory and Retrieval Layer

The memory engine encodes sentences using all-MiniLM-L6-v2 and persists vectors in local ChromaDB collections. Retrieval executes nearest-neighbor matching to select candidate memories for response synthesis.

4.5 Persistence and State

The repository includes persisted assets under a local database directory, including vector storage and model history JSON files. This enables session continuity and model reload without mandatory cloud services.

4.6 Architecture Strengths and Constraints

- Strength: local-first path supports sensitive datasets.
- Strength: stack separation simplifies incremental upgrades.
- Constraint: single-host deployment can bottleneck concurrency.
- Constraint: quality depends on upstream data hygiene and diversity.

- Constraint: without robust confidence calibration, retrieval may return weak matches under sparse corpora.

Chapter 5: Data Pipeline and Training Workflow

CHRONUS training flow is designed to be transparent and reproducible. Every stage, from raw input to indexed memory, is deterministic enough for debugging and auditable enough for sensitive usage.

5.1 Input Acquisition

User-provided files are uploaded to a controlled local directory. The workflow includes data sufficiency checks before training to prevent low-quality or unstable model formation.

5.2 Parsing and Normalization

Text extraction modules parse supported documents into canonical plain-text form.

Normalization handles whitespace cleanup, malformed characters, and basic length constraints to improve embedding behavior.

5.3 Segmentation Strategy

The current implementation uses regex-based sentence splitting for runtime compatibility and portability. Each resulting sentence becomes an addressable memory candidate with lightweight metadata.

5.4 Embedding

Sentence units are transformed to dense vectors with all-MiniLM-L6-v2. This model is selected for stable semantic performance under local execution budgets and practical latency requirements.

5.5 Indexing and Persistence

Embeddings and source text are persisted in ChromaDB, enabling repeated query-time retrieval without re-encoding the entire corpus. Model metadata is tracked for resume and comparison workflows.

5.6 Interview Augmentation

A structured interview module captures high-value dimensions such as relationships, values, habits, and communication style. These responses compensate for sparse documents and improve personalization continuity.

5.7 Failure Modes and Safeguards

- Sparse data mode: low sentence volume yields repetitive or generic outputs.

- Noisy corpus mode: contradictory sources degrade persona coherence.
- Off-topic retrieval mode: weak nearest matches can trigger irrelevant responses.
- Safeguard: enforce data sufficiency checks and low-confidence fallbacks.
- Safeguard: maintain source trace logging for post-hoc review.

Chapter 6: Implementation and Engineering Details

This chapter maps design intent to implemented artifacts in the repository. The goal is implementation realism, not conceptual abstraction.

6.1 API Surface

Endpoint	Purpose
GET /health	Service availability check.
POST /upload	Multi-file ingestion.
GET /files	List uploaded files.
DELETE /files/<filename>	Remove uploaded file.
GET /check-data	Data sufficiency diagnostics.
POST /train	Train memory model from uploaded corpus.
GET /interview/next	Fetch next unanswered interview prompt.
POST /interview/answer	Store interview answer with contributor metadata.
GET /interview/progress	Report interview completion.
GET /pretrained	List available pretrained personas.
POST /pretrained/load	Load selected pretrained persona.
POST /chat	Generate grounded response for user query.

6.2 Frontend Interaction Flow

The UI drives users through upload, data sufficiency validation, optional interview completion, model training, and conversational querying. This flow reduces user error and aligns expectations before inference.

6.3 Pretrained Persona Mechanics

Repository state indicates a configurable pretrained persona mechanism. While one active profile is persisted at runtime, architecture supports extension to larger persona catalogs through standardized memory payloads and style descriptors.

6.4 Logging, Errors, and Recovery

The backend returns explicit HTTP error objects for malformed requests and runtime exceptions. Production hardening should include typed error contracts, request correlation IDs, and security-classified logging.

6.5 Configuration and Environment

Environment-based port configuration, local data paths, and modular script entry points support straightforward local deployment. Future cloud deployment should preserve local processing as a first-class option.

6.6 Performance Considerations

- Embedding cost scales with sentence count; training can be batched.
- Retrieval latency scales with index size and ANN parameters.
- Chat responsiveness depends on retrieval + composition + network overhead.
- Cold start can be reduced via model preloading and warmed indexes.
- Background indexing can improve perceived UX for large uploads.

Chapter 7: Evaluation Methodology and Results

Evaluation combines retrieval metrics, response quality assessments, and system behavior diagnostics. Because CHRONUS is both technical and human-centered, quantitative and qualitative analysis are both required.

7.1 Experimental Design

A controlled query set is prepared from held-out source passages and interview narratives. Each query has one or more expected supporting memory units. The system outputs are compared against expected units and expert human judgment.

7.2 Retrieval Metrics

- Precision@1: top result relevance.
- Precision@3: relevance coverage in top-three retrieval.
- MRR: ranking quality across queries.
- Coverage: percentage of queries with at least one acceptable supporting memory.
- Drift rate: proportion of responses with low topical alignment.

7.3 Response Quality Rubric

Dimension	Definition	Scale
Relevance	How directly the response addresses the user query.	1-5
Groundedness	Degree to which answer aligns with retrievable evidence.	1-5
Persona Consistency	Stability of tone and values over multiple turns.	1-5
Clarity	Ease of understanding and conversational readability.	1-5
Safety	Absence of harmful or manipulative wording.	1-5

7.4 Observational Findings

In pilot runs, retrieval-grounded answers show stronger topical precision and lower fabrication incidence than unconstrained prompting baselines. Structured interview augmentation improves emotional-context answers where documentary sources are thin. Failure cases are concentrated in ambiguous questions and sparse-memory zones.

7.5 Threats to Validity

- Small evaluation cohorts can overstate performance confidence.
- Source quality variability affects benchmark fairness.
- Human raters may differ in judging persona authenticity.
- Ground truth for subjective memories is inherently probabilistic.

Chapter 8: Ethics, Privacy, and Governance

Grief technology demands stronger safeguards than conventional chat systems. The emotional vulnerability of users and the identity rights of represented persons require explicit technical and procedural controls.

8.1 Consent Architecture

- Explicit consent required before ingesting personal archives.
- Role-based permission model for family contributors.
- Consent receipts stored with timestamp and scope.
- Revocation path with hard-delete guarantees.

8.2 Data Protection

- Local storage by default for corpus and indexes.
- Encryption-at-rest for archives and metadata.
- Transport encryption for any remote service calls.
- Minimal retention policy for temporary processing files.
- Audit logs for admin-sensitive actions.

8.3 Psychological Safety

The system should avoid deterministic claims about consciousness or afterlife continuation. UX messaging must clearly frame outputs as memory-grounded simulation. Escalation cues should direct users toward human support in distress scenarios.

8.4 Misuse Mitigation

- Prevent unauthorized identity cloning.
- Restrict political or financial manipulation scenarios.
- Rate-limit high-risk interactions.
- Use forensic watermarking for generated audio/video outputs in future versions.
- Deploy abuse reporting and model lock workflows.

8.5 Compliance Readiness

A production-grade rollout should map controls to applicable data protection and AI governance frameworks, including purpose limitation, deletion rights, explainability disclosures, and auditable incident handling.

Chapter 9: Productization and Deployment Strategy

To transition from prototype to reliable product, CHRONUS needs structured roadmap execution across platform engineering, model governance, and user research.

9.1 Roadmap Phases

Phase	Focus	Primary Deliverables
Phase I	Core Hardening	Typed APIs, observability, robust ingestion tests, confidence fallbacks
Phase II	Human-Centered UX	Consent flows, grief-safe language layers, onboarding guidance
Phase III	Multimodal Expansion	Voice pipeline, optional image/video avatar interfaces
Phase IV	Scalable Operations	Containerization, secure sync options, role-based governance

9.2 Engineering Backlog Priorities

- Introduce retrieval confidence scoring and explainability metadata in response payloads.
- Add benchmark harness for repeatable quality regression testing.
- Implement secure archive encryption and key-rotation policy.
- Add structured model cards for every pretrained and custom persona.
- Prepare plugin interfaces for local and cloud TTS backends.

9.3 Deployment Models

Three deployment modes are recommended: single-device private deployment, family LAN deployment with shared governance, and controlled cloud relay mode for geographically distributed family access with strict data contracts.

9.4 Cost and Feasibility View

A local-first semantic retrieval pipeline remains economically favorable for student and family contexts because it avoids per-token generation dependence for every query. Cost growth is mostly storage and occasional model compute, not perpetual cloud inference burn.

Chapter 10: Conclusion and Future Scope

CHRONUS demonstrates that memory-grounded conversational simulation is technically achievable with a practical full-stack architecture and local vector retrieval workflow. The repository evidence confirms a functioning baseline for data ingestion, training, interview augmentation, and chat interaction.

The key insight is that grounded retrieval is not just a technical optimization; it is an integrity requirement for identity-sensitive AI. In grief contexts, confidence, traceability, and consent are more important than stylistic fluency alone.

10.1 Confirmed Outcomes

- A reproducible local training-to-chat pipeline is operational.
- Semantic retrieval supports meaningful personal Q&A behavior.
- Structured interviews enrich sparse textual corpora.
- Architecture is modular enough for staged capability expansion.

10.2 Future Scope

- Real-time voice synthesis with streaming playback.
- Persona confidence scoring and source-citation overlays in chat UI.
- Multilingual memory retrieval and response support.
- Clinically informed grief-safety protocol integration.
- Optional avatar channels with strict ethical watermarking.
- Federated family contribution workflows with permission controls.
- Longitudinal model updating from new memories over time.

Nomenclature

Symbol / Term	Meaning
AI	Artificial Intelligence
NLP	Natural Language Processing
SBERT	Sentence-BERT sentence embedding framework
TTS	Text-to-Speech synthesis
RAG	Retrieval-Augmented Generation
LLM	Large Language Model
ANN	Approximate Nearest Neighbor
API	Application Programming Interface
$\cos(q,d)$	Cosine similarity between query and document vectors
d	Embedding dimensionality
k	Number of retrieved nearest neighbors
θ	Similarity threshold parameter

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Appendix A: Detailed Chapter Expansions

A.1 Extended Use Cases

CHRONUS can serve bereaved families, oral-history archives, alumni storytelling projects, and educational simulations when strict provenance boundaries are retained. In bereavement, the system should prioritize gentle and non-authoritative phrasing. In heritage contexts, the same architecture can preserve regional languages and oral micro-histories. In education, historical persona modules should disclose interpretive uncertainty and source coverage clearly.

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A.2 Source Curation Protocol

Source curation directly controls model behavior. Documents should be tagged by period, confidence, authorship certainty, and emotional context. Contradictory evidence should not be removed by default; it should be tracked with context metadata. Curators need lightweight tooling to mark uncertain passages and prohibit use of private or legally restricted content.

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A.3 Retrieval Confidence Design

A robust production response should include confidence metadata with human-readable explanation. Example: High confidence when top match margin exceeds threshold and semantic neighborhood is coherent; moderate confidence when matches are topically related but semantically dispersed; low confidence when nearest vectors are weak. Low confidence should trigger clarifying questions rather than speculative responses.

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A.4 Safety Language Templates

Safety templates are required for high-impact prompts, including legal advice, medical advice, severe emotional distress, and requests for certainty about unknowable claims. Templates should acknowledge user emotion, avoid definitive prescriptions, and suggest responsible human support pathways where appropriate.

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A.5 Persona Drift Monitoring

Long conversations can produce persona drift even in grounded systems due to repeated paraphrase accumulation. Drift can be detected by periodic consistency checks against anchor statements from core memories and values. If drift exceeds tolerance, the system can re-anchor response style and disclose the recalibration to users transparently.

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Appendix B: Proposed Advanced Features

- Temporal memory graphs linking events, people, and emotional state over time.
- Cross-modal retrieval: align text memories with photos, audio clips, and video timestamps.
- Family co-authoring with conflict-aware merge workflows for memory contributions.
- Explainable retrieval cards in UI showing why a passage was selected.
- Adaptive interview prompts generated from corpus blind spots.
- On-device speaker adaptation for privacy-preserving voice continuity.
- Secure legacy vault export for legal archival and intergenerational transfer.

Appendix C: Project Management Artifacts

Risk register (summary):

Risk	Impact	Likelihood	Mitigation
Unauthorized data ingestion	High	Medium	Strict consent and access controls
Hallucinated responses	High	Medium	Grounded retrieval and fallback templates
Sparse training corpus	Medium	High	Interview augmentation and sufficiency checks
Operational	Medium	Low	Health checks and

downtime			restart policies
Trust erosion from unclear UX	High	Medium	Transparent simulation disclosures

Appendix D: Extended Discussion Text

The social acceptance of memory simulation systems will depend less on novelty and more on trust architecture. Users must understand what the system can and cannot represent.

Engineering teams should avoid marketing language that implies literal consciousness transfer. Instead, systems should frame themselves as structured, evidence-grounded mirrors of documented memory. This framing protects users from false expectations and guides ethical product evolution. From a technical perspective, the most durable progress pathway is incremental: first robust retrieval and governance, then richer generation, then multimodal embodiment. Skipping foundational controls in pursuit of realism can cause disproportionate harm in grief contexts. Therefore CHRONUS should prioritize reproducibility, auditability, and reversible operations as first-class engineering outcomes.

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